

CSCI 7000

Current Topics in Computer Science

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (30.6 / 100)	26.5 / 100

26

CURRENT AVG SCORE
across 6 job roles

68

POTENTIAL SCORE
with recommended additions

+54

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Covers research topics of current interest in computer science that do not fall into a standard subarea.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	30.6	76.2	+45.6	HIGH
AI Engineer	29.9	69.2	+39.3	HIGH
Data Scientist	28.1	81.7	+53.6	HIGH
MLOps Engineer	26.9	58.2	+31.4	HIGH
Machine Learning Engineer	24.5	62.0	+37.4	HIGH
ML Engineer	19.0	57.8	+38.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Deep Learning

2 of 6 roles

cnn

training

Experiment Tracking 1 of 6 roles

tracking

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

MEDIUM

Consider Deep Learning content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: cnn, training.
- 3

LOW

Optional: Experiment Tracking — would broaden appeal to one additional role. Candidate terms: tracking.

GAP KEYWORDS BY ROLE

Applied Scientist	30.6 → 76.2 (+45.6 pts)
literature interdisciplinary problems models department data learning systems	
AI Engineer	29.9 → 69.2 (+39.3 pts)
perception data robotics models learning systems based problems	
Data Scientist	28.1 → 81.7 (+53.6 pts)
definition data sets presentation social problems approaches	
MLOps Engineer	26.9 → 58.2 (+31.4 pts)
detection data systems learning machine tracking image training	
Machine Learning Engineer	24.5 → 62.0 (+37.4 pts)
data web sets learning tools exploratory design development	
ML Engineer	19.0 → 57.8 (+38.8 pts)
cnn data learning machine systems models world problems	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.